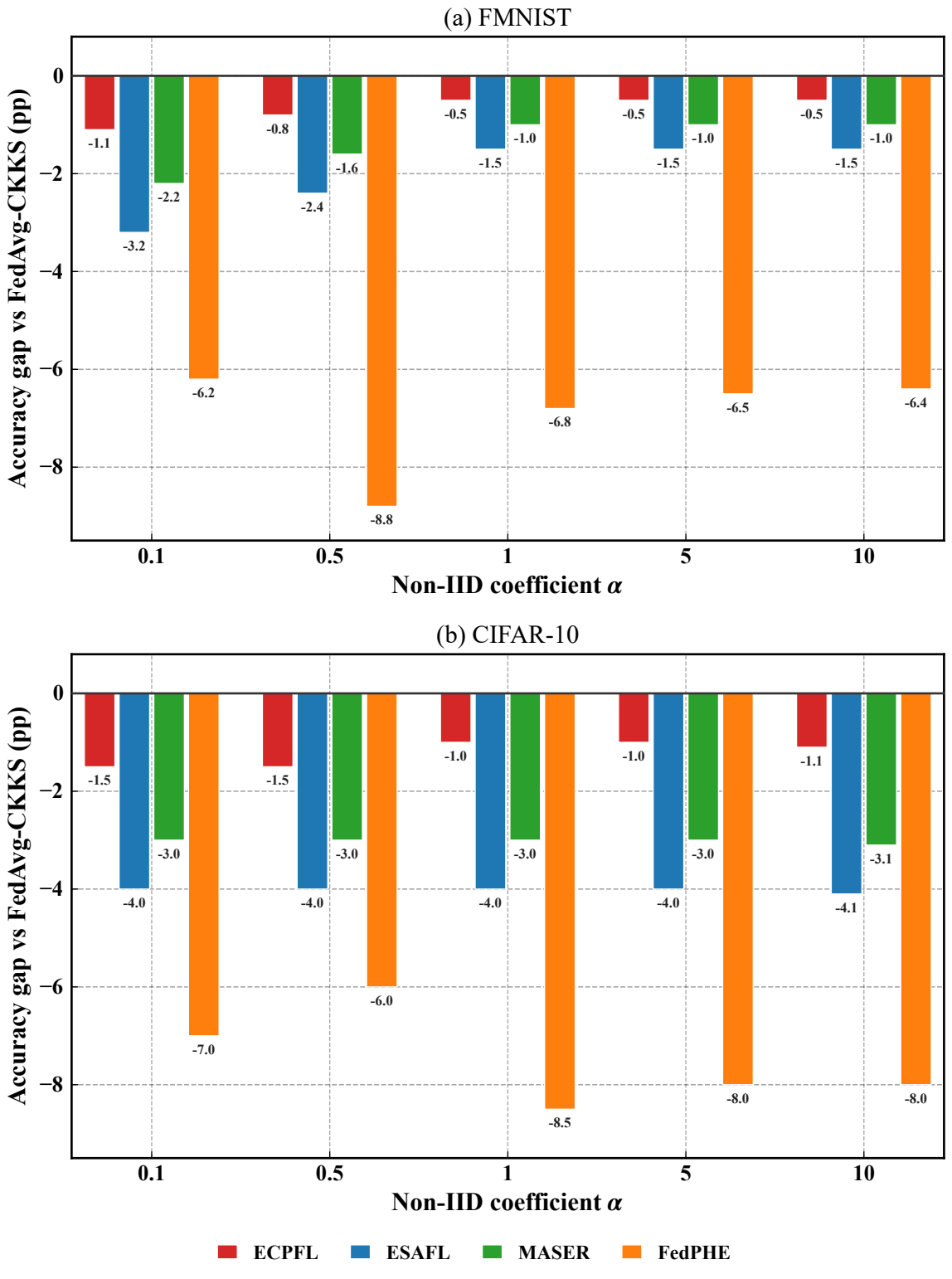
(a) FMNIST
Accuracy gap vs FedAvg-CKKS (pp)
0
−2
−4
−6
−8
-1.1
-3.2
-2.2
-6.2
-0.8
-2.4
-1.6
-8.8
-0.5
-1.5
-1.0
-6.8
-0.5
-1.5
-1.0
-6.5
-0.5
-1.5
-1.0
-6.4
0.1
0.5
1
5
10
Non-IID coefficient α
(b) CIFAR-10
Accuracy gap vs FedAvg-CKKS (pp)
0
−2
−4
−6
−8
-1.5
-4.0
-3.0
-7.0
-1.5
-4.0
-3.0
-6.0
-1.0
-4.0
-3.0
-8.5
-1.0
-4.0
-3.0
-8.0
-1.1
-4.1
-3.1
-8.0
0.1
0.5
1
5
10
Non-IID coefficient α
ECPFL
ESAFL
MASER
FedPHE